

Cho $X, U, V \sim \text{Uniform}(0,1)$ độc lập. Biết:

$$E[X] = E[U] = E[V] = \frac{1}{2}, \text{Var}(X) = \text{Var}(U) = \text{Var}(V) = \frac{1}{12}$$

Đặt $Y = X + U, Z = U + V$. và vector ngẫu nhiên

$$W = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

Kỳ vọng: $E(W) = \begin{bmatrix} E[X] \\ E[X+U] \\ E[U+V] \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1 \\ 1 \end{bmatrix}$

Ma trận hiệp phương sai $\Sigma = \text{Cov}(W)$

Tính từng thành phần: $\text{Var}(X) = \frac{1}{12}$ ~~$\text{Var}(Y)$~~

$$\text{Var}(Y) = \text{Var}(X) + \text{Var}(U) = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}$$

$$\text{Var}(Z) = \text{Var}(U) + \text{Var}(V) = \frac{1}{6}$$

$$\text{Cov}(X, Y) = \text{Cov}(X, X+U) = \text{Var}(X) = \frac{1}{12}$$

$$\text{Cov}(X, Z) = \text{Cov}(X, U+V) = 0 \quad (\text{vì } X \perp U, V)$$

$$\text{Cov}(Y, Z) = \text{Cov}(X+U, U+V) = \text{Var}(U) = \frac{1}{12}$$

Do đó: $\Sigma = \begin{bmatrix} \frac{1}{12} & \frac{1}{12} & 0 \\ \frac{1}{12} & \frac{1}{6} & \frac{1}{12} \\ 0 & \frac{1}{12} & \frac{1}{6} \end{bmatrix}$

- Ma trận hệ số tỷ quan R

- Độ lệch chuẩn: $\sigma_x = \sqrt{\frac{1}{12}}$, $\sigma_y = \sigma_z = \sqrt{\frac{1}{6}}$

Các hệ số: $\rho_{x,y} = \frac{\frac{1}{12}}{\sqrt{\frac{1}{12} \cdot \frac{1}{6}}} = \frac{\sqrt{2}}{2}$

$\rho_{x,z} = 0$, $\rho_{y,z} = \frac{1/12}{1/6} = \frac{1}{2}$

Vậy $R = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ \frac{1}{2} & 1 & \frac{1}{2} \\ 0 & \frac{1}{2} & 1 \end{bmatrix}$

$$f_{X_1, X_2, X_3, X_4}(x_1, x_2, x_3, x_4) = 9 \times 10^{-12} e^{-0.001x_1} e^{-0.002x_2} e^{-0.0015x_3} e^{-0.003x_4}$$

$$= 9 \cdot 10^{-12} e^{-0.001x_1} \cdot e^{-0.002x_2} \cdot e^{-0.0015x_3} \cdot e^{-0.003x_4}$$

f) $P(X_1 > 1000, X_2 > 1000, X_3 > 1000, X_4 > 1000)$

$$= \int_{1000}^{\infty} \int_{1000}^{\infty} \int_{1000}^{\infty} \int_{1000}^{\infty} f(x_1, x_2, x_3, x_4) dx_1 dx_2 dx_3 dx_4 \quad a)$$

~~$f_{X_1, X_2, X_3, X_4}(x_1, x_2, x_3, x_4) = 9 \cdot 10^{-12} e^{-0.001x_1} e^{-0.002x_2} e^{-0.0015x_3} e^{-0.003x_4}$~~

$$P. = 9 \cdot 10^{-12} \prod_{i=1}^4 \int_{1000}^{\infty} e^{-a_i x_i} dx_i$$

$$= 9 \cdot 10^{-12} \prod_{i=1}^4 \left(-\frac{1}{a_i} e^{-a_i x_i} \Big|_{1000}^{\infty} \right)$$

$$= 9 \cdot 10^{-12} \prod_{i=1}^4 \left(\frac{1}{a_i} e^{-a_i \cdot 1000} \right)$$

$$= 9 \cdot 10^{-12} \cdot \frac{1}{0.001 \cdot 0.002 \cdot 0.0015 \cdot 0.003} e^{1000(-0.001-0.002-0.0015-0.003)}$$

$$= \underline{9 \cdot 10^{-12}} \cdot e^{-7.5}$$